

1 Basic Knowledge & SVCDE Methods for ODE

1.1 Linear Algebra || Calculus || SVCDE Method for ODE

Matrix Calculate:  $\det(kA) = k^n \det(A)$  ||  $\det(A^{-1}) = \det(A)^{-1}$  ||  $(AB)^* = B^* A^*$  ||  $\det(A) = \det(A^T)$ .

- Eigenvalue  $\lambda$  & Eigenvector  $\mathbf{x}$ : Def:  $A\mathbf{x} = \lambda\mathbf{x}$ . || Method:  ${}^1 \det(A - \lambda I) = 0 \rightarrow \lambda$ .  ${}^2 (A - \lambda I)\mathbf{x} = 0 \rightarrow \mathbf{x}$ .
- Arithmetic Manipulate (AM)  $\lambda$ :  $\lambda$  重复出现的次数.  $\geq$  Geometric Manipulate (GM) of  $L$ :  $\# \mathbf{x}$  under  $\lambda$ .
- Generalised Eigenvectors: If  $AM > GM$ : Using  $\xi_\lambda = (A - \lambda I)\eta_1, \eta_{n-1} = (A - \lambda I)\eta_n$  for each  $\xi_\lambda$ .  $\eta_j$  are generalised eigenvectors.

Some Defs:  $\nabla F = (\partial_x F, \partial_y F, \partial_z F, \dots)$  || Simply Connected: 没有洞, 相互连接的区域. ||  $\partial D$ : 区域 D 的边界.

- Method of Integrating Factors:  $y' + p(x)y = q(x)$ .  $\Rightarrow$  Integrating Factor:  $\mu = e^{\int p(x)dx}$ ,  $\mu y = \int q(x) \cdot \mu dx$
- Homogeneous 2nd ODE form:  $ay'' + by' + cy = 0 \Rightarrow$  Characteristic  $ar^2 + br + c = 0$ , Roots:  $r_1, r_2$  ps:  $y'' \pm \lambda y = 0$  见附录
- ${}^1 \Delta > 0, y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$ ;  ${}^2 \Delta = 0, y = (C_1 + C_2 x)e^{rx}$ ;  ${}^3 \Delta < 0, y = e^{\alpha x}[C_1 \cos(\beta x) + C_2 \sin(\beta x)]$  ps:  $r_1, r_2 = \alpha \pm \beta i$
- non-Homogeneous 2nd ODE form:  $ay'' + by' + cy = g(x) \Rightarrow y = y_h + y_p$  ps:  $y_h$  is sol for  $g(x) = 0$ .
- Method of Undetermined Coefficients: Let  $A_n(x) = \sum_{i=0}^n a_i x^i$ ,  $B_n(x) = \sum_{i=0}^n b_i x^i$ ,  $P_n(x) = \sum_{i=0}^n p_i x^i$ .

- 1. If  $g(x) = P_n(x) \Rightarrow y_p = x^S A_n(x)$ ,  $S := 0, (0 \neq r_1, r_2); 1, (0 = r_1 \text{ or } r_2); 2, (0 = r_1 = r_2)$ .
- 2. If  $g(x) = M e^{kx} \Rightarrow y_p = C x^S e^{kx}$ ,  $S := 0, (k \neq r_1, r_2); 1, (k = r_1 \text{ or } r_2); 2, (k = r_1 = r_2)$ .
- 3. If  $g(x) = M \cos(kx) + N \sin(kx) \Rightarrow y_p = x^S (C_1 \cos(kx) + C_2 \sin(kx))$ ,  $S := 0, (r_1, r_2 \neq \pm ik); 1, (r_1, r_2 = \pm ik)$ .
- 4. If  $g(x)$  = product from 1,2,3.  $\Rightarrow y_p$  = corresponding product of  $y_p$  ( $x^S$  只乘一次),  $S :=$  令  $y_p$  与  $y_c$  没有相同项. ( $\leq 2$ )

Euler Equation: General:  $a_n x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \dots + a_0 y = 0 \Rightarrow$  Seek Solution as:  $y = x^\alpha \Rightarrow$  solution:  $y = \sum_i c_i x^{\alpha_i}$

2.1 Homogeneous

Principle of Superposition: If  $y_i$  are solutions  $\Rightarrow \sum_i c_i y_i$  also. ||  $S := \{y | y' = Py\}$  is a linear space,  $\dim(S) = n$  if  $P_{ij}$  continuous.

Fundamental Set of Solutions (FSoS): If  $\{y^{(j)}(t)\}$  linearly independent at  $t \in I$ , it is FSoS in  $I$ .

General Solution: If  $\{y^{(j)}(t)\}_{j=1}^n$  is FSoS (Linearly Independent)  $\Rightarrow$  General solution:  $y = \sum_{j=1}^n c_j y_j$

2.1.1 Eigenvalue/vector Method (A constant)

AM=GM|Real  $\lambda$ : general solution:  $y = \sum_j c_j e^{\lambda_j t} \xi_j$ .

AM=GM|Complex  $\lambda$ : (ctd.) For  $\lambda, \bar{\lambda} \in \mathbb{C} \Rightarrow e^{\lambda t} \xi = \mathbf{u} + vi \Rightarrow$  Independent sol:  $\mathbf{u}, \mathbf{v} \Rightarrow$  General solution:  $\mathbf{y} = c_1 \mathbf{u} + c_2 \mathbf{v} + \dots$

AM>GM: (ctd.) (For each  $\xi_\lambda$ , under  $\lambda$ )  $\mathbf{y}^{(1)} = e^{\lambda t} \xi_\lambda$   $\mathbf{y}^{(2)} = e^{\lambda t} (t \xi_\lambda + \eta_1)$   $\mathbf{y}^{(3)} = e^{\lambda t} (\frac{t^2}{2!} \xi_\lambda + t \eta_2 + \eta_1) \dots$  类似泰勒展开

2.1.2 Matrix Method | Exponential

Matrix Method:  $\mathbf{y}(t) = e^{At} \mathbf{y}(0)$  (分两种方法: 1. 直接展开; 2. 通过一些已有 property 计算. ps:  $e^{At} = \Psi$ )

- Property|AM=GM:  $e^{At} = T e^{Dt} T^{-1}$ ,  $T = [\xi_1, \dots, \xi_n]$ ,  $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ ,  $e^{Dt} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})$
- Property|AM>GM:  $e^{At} = T e^{Jt} T^{-1}$ ,  $T = [\lambda_1, \dots, \lambda_i, \eta_1^{(1)}, \dots, \lambda_j, \eta_1^{(j)}, \dots]$ ,  $J = T^{-1} A T$  (Jordan form),  $e^{Jt}$  = 见右边

Diagonalisation Method: (Ctd.) Let  $\mathbf{y} = T\mathbf{x} \Rightarrow \mathbf{x}' = J\mathbf{x}$  [or  $\mathbf{x}' = D\mathbf{x}$ ] 按前面方法解

2.1.3 Matrix Method | Fundamental Matrix

Fundamental matrix  $\Psi$ : If  $\{y^{(j)}(t)\}$  is FSoS.  $\Rightarrow \Psi(t) = [y^{(1)}(t), \dots, y^{(n)}(t)]$ .

· Property:  $\Psi'(t) = P(t)\Psi(t)$  (It's a sol). ||  $\det[\Psi(t)] = W(t) \neq 0$  || general solution:  $\mathbf{y} = \Psi \mathbf{c}$ ,  $\mathbf{c} = \Psi^{-1}(t_0) \mathbf{y}_0$

· Relation with  $e^{At}$ :  $e^{At} = \Psi(t)$  with  $\Psi(0) = \mathbb{I}$  ||  $e^{At} = \Psi(t) \Psi^{-1}(0)$

2.2 non-Homogeneous

Theorem: Solution of  $(*)$ :  $\mathbf{y} = \sum_i c_i y^{(i)} + \mathbf{y}_p$ , where  $y^{(i)}$  is solution of it's Homogeneous.

通常会让特解中积分后的常数 c=0

Variation of Parameters:  $\mathbf{y} = \mathbf{y}_g + \mathbf{y}_p$ ;  $\mathbf{y}_g = \Psi(t) \Psi^{-1}(t_0) \mathbf{y}(t_0)$ ;  $\mathbf{y}_p = \Psi(t) \mathbf{u}(t)$ ,  $\mathbf{u}(t) = \int_{t_0}^t \Psi^{-1}(s) \mathbf{g}(s) ds$ . 经常  $\Psi(t) \mathbf{u}' = \mathbf{g}$  来求  $\mathbf{u}$

Matrix methods|Diagonalisation:  ${}^1$  Let  $\mathbf{y} = T\mathbf{x}$   ${}^2 (*) \Rightarrow \mathbf{x}' = D\mathbf{x} + T^{-1} \mathbf{g} = D\mathbf{x} + \mathbf{h}$   ${}^3$  直接计算;  $x_i = c_i e^{r_i t} + e^{r_i t} \int_{t_0}^t e^{-r_i s} h_i(s) ds$

Undetermined Coefficients: when  $\mathbf{g}(t)$  is polynomials, or real/complex exponential. Let  $P_n(t) = \sum_{i=0}^n \mathbf{p}_i t^i$ ,  $A_n(t) = \sum_{i=0}^n \mathbf{a}_i t^i$

${}^1$  构造  $y_p$  与  $\mathbf{g}$  结构相匹配 (e.g. If  $\mathbf{g} = P_n(t) \Rightarrow y_p = A_n(t)$ ; If  $\mathbf{g} = c e^{kt} \Rightarrow y_p = a e^{kt}$ )  ${}^2$  特殊: If  $\mathbf{g} = c e^{\lambda t} \lambda$  is eigenvalue with  $AM = n \Rightarrow y_p = e^{\lambda t} \sum_{i=0}^n \mathbf{a}_i t^i$   ${}^3$  带入原方程得未知系数

2.3 Geometrical Interpretation & Linear Approximation

\* 此节中, Consider nonlinear autonomous system:  $\mathbf{x}' = F(x, y)$ ,  $y' = G(x, y)$  (\*\*)

Stable: Critical Point  $\mathbf{x}_0$  is stable iff  $\forall \epsilon > 0, \exists \delta > 0$  s.t.  $\forall$  sols  $\mathbf{x}(t)$  with  $\|\mathbf{x}(0) - \mathbf{x}_0\| < \delta$ , then  $\|\mathbf{x}(t) - \mathbf{x}_0\| < \epsilon, \forall t \geq 0$  || 直观理解: 开始近  $\rightarrow$  保持近.

Attracting: Critical point  $\mathbf{x}_0$  is attracting iff  $\exists \delta > 0$  s.t.  $\forall$  sols  $\mathbf{x}(t)$  with  $\|\mathbf{x}(0) - \mathbf{x}_0\| < \delta$  satisfies:  $\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{x}_0$  || 直观理解: 开始近  $\rightarrow$  不断趋向.

Asymptotically Stable: Critical point  $\mathbf{x}_0$  is asystab iff  ${}^1$  Stable,  ${}^2$  Attracting.

Theorem: If  $\mathbf{x}_*(t) = (x_*, y_*)$  s.t.  $F(\mathbf{x}_*), G(\mathbf{x}_*) \neq 0 \Rightarrow \exists H$  s.t.  $H(x, y) = (\tilde{x}, \tilde{y}), \tilde{x}' = 1, \tilde{y}' = 0$  H is a smooth change of variable function in neighbourhood of  $\mathbf{x}_*$ .

Non-Linear  $\rightarrow$  Linear: For system (\*\*), assume  $\mathbf{x}_0 = (x_0, y_0)$  is critical point.  $\Rightarrow \mathbf{u} = \mathbf{x} - \mathbf{x}_0$ , then  $\mathbf{u}' \approx A \mathbf{u}$  Linear;  $A = \begin{pmatrix} \partial_x F(\mathbf{x}_0) & \partial_y F(\mathbf{x}_0) \\ \partial_x G(\mathbf{x}_0) & \partial_y G(\mathbf{x}_0) \end{pmatrix}$  jacobian

2.4 Classification of Points

\* 此节中, Consider linear system:  $\mathbf{x}' = A\mathbf{x}$ . Critical Point(s): when  $\mathbf{x}' = A\mathbf{x} = 0$

Classification of Critical Points at  $(x_0, y_0)$  (Linear):  $r_1, r_2$  be sol of  $\det(A - \lambda I) = 0$ . ||  $\mathbb{C} : r = \lambda \pm i\mu (\mu > 0)$  || If A constant, write sol:  $\mathbf{x} = c_1 e^{r_1 t} \xi_1 + c_2 e^{r_2 t} \xi_2$  ||  $GM = 1: \mathbf{x} = c_1 e^{rt} \xi + c_2 e^{rt} (t \xi + \eta)$

| ℝ/ℂ                                                                                                                                                                                                              | Condition    Stability                     | Type    Name                       | Phase Plane Description                                                                   | Other                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ℝ                                                                                                                                                                                                                | $r_1 < r_2 < 0$    asystab                 | N    NSk                           | 向原点, $\xi_2$ 直线, $\xi_1$ 曲线, 和 $\xi_1$ 周围 $y = \pm x^3$                                   | $c_2 \neq 0, t \rightarrow \infty: \xi_2$ 主导方向; $c_2 = 0, t \rightarrow \infty: \xi_1$ 主导方向                                                                                                                                                                                                                 | PS:<br>N = Node<br>PN = Proper Node<br>IN = Improper<br>or: Degenerate Node<br>SP = Saddle Point<br>SpP = spiral point<br>or: Focus Point<br>C = Center<br>NSk = Nodal Sink<br>NSo = Nodal Source |
|                                                                                                                                                                                                                  | $r_1 > r_2 > 0$    unstable                | N    NSo                           | 原点向外, $\xi_2$ 直线, $\xi_1$ 曲线, 和 $\xi_1$ 周围 $y = \pm x^3$                                  | $c_1 \neq 0, t \rightarrow \infty: \xi_1$ 主导方向; $c_1 = 0, t \rightarrow \infty: \xi_2$ 主导方向                                                                                                                                                                                                                 |                                                                                                                                                                                                   |
|                                                                                                                                                                                                                  | $r_1 > 0 > r_2$    unstable                | SP    SP                           | $t \rightarrow \infty, \xi_1$ 从原点向外, $\xi_2$ 从外向原点<br>and: 像 $y = \pm \frac{1}{x}$ , 同进同出 | $t \rightarrow \pm \infty:  \mathbf{x}  \rightarrow \infty$ ; $t \rightarrow \infty: c_1, c_2 \neq 0,  \mathbf{x}  \rightarrow \infty, \xi_1$ 主导;<br>$t \rightarrow \infty: c_2 = 0,  \mathbf{x}  \rightarrow \infty, \xi_1$ 主导; $t \rightarrow \infty: c_1 = 0,  \mathbf{x}  \rightarrow \infty, \xi_2$ 主导 |                                                                                                                                                                                                   |
|                                                                                                                                                                                                                  | $r_1 = r_2 < 0$ , GM=2    asystab          | PN    PN or Stable Star            | 直线 向原点                                                                                    | 直线, $u_1 / u_2$ is $t$ independent                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                   |
|                                                                                                                                                                                                                  | $r_1 = r_2 > 0$ , GM=2    unstable         | PN    PN or Unstable Star          | 直线 从原点向外                                                                                  | 直线, $u_1 / u_2$ is $t$ independent                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                   |
|                                                                                                                                                                                                                  | $r_1 = r_2 < 0$ , GM=1    asystab          | IN (AL-Type: SpP)    IN (Stable)   | S 曲线, 向原点                                                                                 | $t \rightarrow \infty,  \mathbf{x}  \rightarrow 0, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2 \xi$ 方向相同                                                                                                                                                                                                                |                                                                                                                                                                                                   |
| ℂ                                                                                                                                                                                                                | $r_1 = r_2 > 0$ , GM=1    unstable         | IN (AL-Type: SpP)    IN (Unstable) | S 曲线, 从原点向外                                                                               | $t \rightarrow \infty,  \mathbf{x}  \rightarrow \infty, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2 \xi$ 方向相同                                                                                                                                                                                                           |                                                                                                                                                                                                   |
|                                                                                                                                                                                                                  | $\lambda \neq 0, \lambda > 0$    unstable  | SpP    Unstable Focus              | 向外椭圆 (elliptical) 螺旋                                                                      | $t \rightarrow \infty,  \mathbf{x}  \rightarrow \infty$ ps: 考虑 $A = (a, b, c, d)$ , 如果 $bc > 0$ , 顺时针, 如果 $bc < 0$ , 逆时针                                                                                                                                                                                    |                                                                                                                                                                                                   |
|                                                                                                                                                                                                                  | $\lambda \neq 0, \lambda < 0$    asystab   | SpP    Stable Focus                | 向内椭圆 (elliptical) 螺旋                                                                      | $t \rightarrow \infty,  \mathbf{x}  \rightarrow 0$ ps: 考虑 $A = (a, b, c, d)$ , 如果 $bc > 0$ , 顺时针, 如果 $bc < 0$ , 逆时针                                                                                                                                                                                         |                                                                                                                                                                                                   |
|                                                                                                                                                                                                                  | $\lambda = 0$    stable (AL:Indeterminate) | C (AL:C or SpP)    C               | 椭圆 (elliptical) 和 半长轴 $\xi$ 实部方向                                                          | Bounded trajectory or $\exists$ Periodic Trajectories                                                                                                                                                                                                                                                       |                                                                                                                                                                                                   |
| AL: Almost Linear Sys.    For Almost Linear systems: $\mathbf{u}' = A\mathbf{u} + \eta$ , 如果 linear Approximation (去掉 $\eta$ ), 按照 Linear 考虑. 如果不去 $\eta$ , 按照去掉的算 $r_1, r_2$ , Type 按照正常 / (AL) 的来.(图像会到 (0,0)) |                                            |                                    |                                                                                           |                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                   |

Some Defs: Separatrix: 将不同区域分开的轨迹; Basin of attraction: 所有趋近 critical point 的轨迹. ps: In any nonlinear ODEs, it's important to determine the basin of attraction.

3 Non-Linear ODE Methods

\* 此节中, Consider two-dimensional autonomous system:  $\mathbf{x}' = F(x, y)$ ,  $y' = G(x, y)$  and  $\mathbf{x} = (x, y)^T$  (#) Lyapunov's: 不需要知道微分方程的确切解 critical value. Poincaré-Bendixson: 寻找 limit cycle

Lyapunov Stability Theorem:  ${}^1$  (#);  ${}^2$  ICP  $\mathbf{x}_0 \in D$ ;  ${}^3$   $\exists$  正定且连续  $E: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, E(\mathbf{x}_0) = 0$ ; ||  ${}^4 \frac{dE}{dt} = \nabla E \cdot \mathbf{v} = \partial_x E \cdot F + \partial_y E \cdot G$  (负 (#) 定) on  $D \setminus \mathbf{x}_0(D) \Rightarrow \mathbf{x}_0$  asp.(stab)

· Lyapunov function: 符合上面 1-4 条件的

Lyapunov Instability Theorem:  ${}^1$  (#);  ${}^2$  ICP  $\mathbf{x}_0 \in D$ ;  ${}^3$   $\exists E: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, \partial_t E$  连续,  $E(\mathbf{x}_0) = 0$ ;  ${}^4$   $\nabla$  领域  $\mathbf{x}_0, \exists \mathbf{x}_1 (E(\mathbf{x}_1) > 0)$ ; ||  ${}^5 \frac{dE}{dt}$  正定 on  $D \Rightarrow \mathbf{x}_0$  unstable

Periodic Solution: Soln  $\mathbf{x}$  has:  $\mathbf{x}(t + T) = \mathbf{x}(t), \forall \mathbf{x}$ . || 闭合曲线  $\Leftrightarrow$  Periodic sol || Periodic Solution + 附近有吸引或排斥  $\Rightarrow$  Limit Cycle || when  $\dot{r} = 0, \dot{\theta} = \text{const}$ . Periodic sol

Limit cycle: Limit cycle  $\Rightarrow$  Periodic Solution || Def  $t \rightarrow \pm \infty, \exists$  non-closed trajectory asymptotes to it. (inside or outside) ps: It's Orbital Stability

- 1. Limit cycle is:  ${}^1$  asystab:  $\forall$  trajectories asy.it;  ${}^2$  semistable:  $\exists$  trajectories 单方向进/出;  ${}^3$  unstable:  $\forall$  trajectories away from it.
- 2. Theorem (If  $\exists$  limit cycle / closed):  ${}^1 F, G$  一阶偏导连续 in simply connected  $D$ ;  ${}^2$  被限环/封闭环  $(*)$  in  $D \Rightarrow {}^1 \exists$  CP in  $(*)$ ;  ${}^2$  If  $\exists$  CP in  $(*)$ , it's not SP.
- 3. Theorem (No limit cycle / closed):  ${}^1 F, G$  一阶偏导连续 in simply connected  $D$ ;  ${}^2 \partial_x F + \partial_y G$  符号不变 on  $D \Rightarrow$  No closed trajectory in  $D$ .

${}^2$  的推论:  ${}^1$  If simple connected  $D$  no CP  $\Rightarrow$  No closed trajectory in  $D$ .  ${}^2$  If simple connected  $D$  has unique CP  $\Rightarrow$  No closed trajectory in  $D$

Poincaré-Bendixson Theorem:  ${}^1 F, G$  一阶偏导连续 in  $D$ ;  ${}^2 \exists D_1 \subset D$  bounded;  ${}^3 R = D_1 \cup \partial D_1$  has NO CP; ||  ${}^4 \exists$  soln  $\mathbf{x} \in R, \forall t \geq t_0 \Rightarrow$

$\Rightarrow {}^1 \exists$  closed trajectory.  ${}^2$  Either  $\mathbf{x}$  spirals towards a Periodic trajectory. or  $\mathbf{x}$  periodic  ${}^3 R$  is not simply connected

Find Trapping Region / Limit Cycle: trapping region: 符合上面定理 2-4 除 CP 的 region. Method: polar coordinate, 计算  $r^2 = x^2 + y^2$  对  $t$  的求导, 寻找  $rr' \geq 0$  or  $rr' \leq 0$

4 Fourier Series

\* BVPs 可能没有解/一个解/无数解; IVP 只会有一个解

4.1 Function Space, Eigenfunctions, Periodic Function and Orthogonal Function

Periodic Functions: Function  $f(x)$  is Periodic iff  $f(x + T) = f(x), \forall x$ . Period: T. Fundamental Period: The smallest possible T. || Property: If  $f, g$  has same period T.  $\Rightarrow af(x) + bg(x)$  also has period T.

Differential Operators: Any linear combination of  $n$ -th derivatives defines a linear map or linear operator. e.g.  $p(x) \frac{d}{dx} + \frac{d^2}{dx^2} : y \mapsto p(x) \frac{dy}{dx} + \frac{d^2 y}{dx^2}$

Eigenvalues & Eigenfunctions: If  $D$  is a differential operator  $\Rightarrow$  If  $Df = \lambda f$ , then  $\lambda$  is Eigenvalue and  $f$  is Eigenfunctions || Property: 对于微分方程:  $Df = \lambda f \Rightarrow$  solution:  $f = \sum c_n f_n$   $f_n$  are Eigenfunctions

Inner Product:  $(f, g) = \int_a^b f(x)g(x)dx$  For funcs  $f, g$  in  $(a, b)$  || Orthogonal:  $(f, g) = 0$  || Square Norm:  $(f, f)$ . || Kronecker delta:  $\delta_{mn} = \{0, m \neq n; 1, m = n$ .

1.2 Linear || (ODE) Uniqueness Theorem || Autonomous

Def of Linear: Let  $L$  去掉  $y_i^{(n)}$  的函数 (e.g.  $L = \frac{d^2}{dx^2} - x$  for  $y'' - xy = x^2$ )  
 $\Rightarrow$  Linear if:  $L[a_1 y_1 + a_2 y_2] = a_1 L[y_1] + a_2 L[y_2]$

Existence & Uniqueness Theorem (Real func):

- ${}^1 y'(t) = F(t, y^T), y(t_0) = y_0$ . Let  $R_t = [t_0 - \delta, t_0 + \delta] \times \prod_j [y_{0j} - \delta_j, y_{0j} + \delta_j]$ .
- ${}^2 F_j, \partial_{y_i} F_j : R_t \rightarrow \mathbb{R}$  are continuous.  $\Rightarrow \exists$  unique solution  $\mathbf{y}$  in  $R_t'$ .

$$R_t' = [t_0 - \varepsilon, t_0 + \varepsilon] \times \prod_j [y_{0j} - \varepsilon_j, y_{0j} + \varepsilon_j] \subseteq R_t.$$

Autonomous: For  $y'(t) = f(t, y^T)$ .  $\partial_x f = 0 \Rightarrow$  autonomous. or:  $f(t, y^T) = f(y^T)$

2 Linear ODEs

$n$ -th ODE  $\rightarrow$  1st ODEs: For  $y^{(n)} = F(t, y, y', \dots, y^{(n-1)})$ .  $\Rightarrow$  Let  $x_i = y^{(i-1)}$ .

Def of Linear 1st ODEs:  $y'(t) = P(t)y(t) + g(t) (*)$ ,  $P$  is matrix-valued function.

Homogeneous:  $g(t) = 0$ .

Existence & Uniqueness Theorem: If  $P_{ij}, g$  of  $(*)$  continuous in  $R, y(t_0) = y_0$

$\Rightarrow \exists$  unique solution  $\mathbf{y}$  in  $R'$   $R' = [t_0 - \varepsilon, t_0 + \varepsilon] \subseteq R = [t_0 - \delta, t_0 + \delta]$

Solve:  $y' = Py$  || If  $P$  constant,  $P = A$  ||  $\lambda, \xi, \eta$  be 特征值/向量/广义特征向量.

Abel's Theorem: If  $W(t_0) \neq 0, t_0 \in I \Rightarrow \forall t \in I, W(t) \neq 0$

Abel's Formula:  $W(t) = W(t_0) e^{\int_{t_0}^t \text{tr}[P(s)] ds}$ ;  $t_0, t \in I$   $P(t)$  is continuous in  $I$ .

Wronskian at  $x$ :  $W[y^{(1)}, \dots, y^{(n)}](t) = \begin{vmatrix} y_1^{(1)}(t) & \dots & y_1^{(n)}(t) \\ \dots & \dots & \dots \\ y_n^{(1)}(t) & \dots & y_n^{(n)}(t) \end{vmatrix}$ .

$W(t) \neq 0, \forall t \Rightarrow \{y^{(j)}\}$  linearly independent. ps: 经常用 Abel Theorem

$$e^{Jt} = e^{\lambda_i t} \begin{pmatrix} 1 & t & \frac{t^2}{2} & \dots & \frac{t^{n-1}}{(n-1)!} \\ 0 & 1 & t & \dots & \frac{t^{n-2}}{(n-2)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix} \text{ ps: (e.g.) } J_3 = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

Exponential of Matrix:  $e^{At} := \sum_{n=0}^{\infty} \frac{(At)^n}{n!} = \mathbb{I} + At + \frac{1}{2!} A^2 t^2 + \dots$   
 $:= \lim_{n \rightarrow \infty} (\mathbb{I} + \frac{At}{n})^n$

· Property:  $e^{A+B} = e^A e^B$  if  $AB = BA$  ||  $e^{A \cdot 0} = \mathbb{I}$  ||  $[e^{At}]' = A e^{At}$

$$\text{Inverse of Matrix: } \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$
$$(e.g.) J_3^{-1} = \frac{1}{\lambda} \begin{pmatrix} 1 & -\frac{1}{\lambda} & \frac{1}{\lambda^2} \\ 0 & 1 & -\frac{1}{\lambda} \\ 0 & 0 & 1 \end{pmatrix}$$

4.2 Fourier Series

**Fourier Series:** Periodic function with  $T = 2L$  ,  $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L}) \right)$  (\*)

· **Euler-Fourier Formulas:** Suppose (\*) converges in the interval  $[-L, L]$ . Then:  $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos(\frac{n\pi x}{L}) dx$      $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin(\frac{n\pi x}{L}) dx$      $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$  (Average Value of  $f$  over  $2L$ )

**Piecewise Continuous:**  $f(x)$  piecewise continuous on  $[a, b]$  iff  $\exists a = x_0 < \dots < x_n = b$  s.t.  $^1 f(x)$  continuous  $\forall x \in (x_{i-1}, x_i)$      $^2 f(x_i^{\pm}) = \lim_{x \rightarrow x_i^{\pm}} f(x)$  finite.

**Fourier Convergence Theorem:** If  $^1 f, f'$  piecewise continuous on  $[-L, L]$ ,  $^2 f(x + 2L) = f(x)$ .  $\Rightarrow$  Fourier Series of  $f$  converges.    **Fourier 和 SL 都不一定逐点收敛, 但一定均方收敛.** (类似泰勒展开)

· Furthermore, Let Fourier Series of  $f$  is  $F$ .     $F(x) \rightarrow f(x)$ ,  $f(x)$  continuous;  $F(x) \rightarrow \frac{f(x^+) + f(x^-)}{2}$ ,  $f(x)$  discontinuous.

**Term-By-Term Integration:** Fourier Series for *piecewise continuous* functions can be integrated term-by-term to Convergent Fourier Series.

**Finite Truncation (Partial Sum):** Finite Truncation of Fourier Series is:  $s_k = \frac{a_0}{2} + \sum_{n=1}^k \left( a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L}) \right)$  || **Error:**  $e_N = s_N - f(x)$

·  $n \rightarrow \infty$ :  $^1$  Continuous points, Fourier approximation of  $f(x)$  is better (points decrease).     $^2$  Discontinuous points, oscillation of Fourier series get smaller.

· **Gibbs Phenomenon:** The Amplitude of the *largest* oscillation (near discontinuous points) does *not decrease*.

4.3 Other Results on Fourier Series

**Extension of a function:**  $f(x)$  know for  $x \in [0, L] \Rightarrow$  Even Ext.:  $h(x) = \{f(x), [0, L]; f(-x), (-L, 0) \}$  || Odd Ext.:  $h(x) = \{f(x), (0, L); 0, x = \pm L, 0; -f(-x), (-L, 0)$

Even Ext:  $a_n = \frac{2}{L} \int_0^L f(x) \cos(\frac{n\pi x}{L}) dx, b_n = 0$  || Odd Ext:  $b_n = \frac{2}{L} \int_0^L f(x) \sin(\frac{n\pi x}{L}) dx, a_n = 0$

**Complex Form of Fourier Series:**  $f(x) = \sum_{-\infty}^{\infty} c_n e^{\frac{i n \pi x}{L}}$  ||  $c_n = \frac{a_n - i b_n}{2}; c_{-n} = \frac{a_n + i b_n}{2}; c_0 = \frac{a_0}{2}, n > 0$  ||  $a_0 = 2c_0; a_n = c_n + c_{-n}; b_n = i(c_n - c_{-n})$

· **Euler-Fourier Formula (Complex):** If (f) of Periodic  $f$ ,  $(T = 2L)$  converges in  $[-L, L] \Rightarrow c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-\frac{i n \pi x}{L}} dx$  ||  $c_{-n} = \overline{c_n}$  if  $f$  real.

**Parseval's Theorem:** If  $^1$  Periodic function  $f$ ;  $^2 f$  has convergent Fourier Series.  $\Rightarrow$  Square Norm  $\langle f, f \rangle = \int_{-L}^L |f(x)|^2 dx = 2L \sum_{n=-\infty}^{\infty} |c_n|^2 = L \left[ \frac{|a_0|^2}{2} + \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2) \right]$  ps: 无限维的勾股定理

5 Linear PDEs by Separation of Variables

**Requirement:** Assume  $u(x, t) = X(x)T(t)!$   $\Rightarrow$  If PDE 能写成类似  $\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} \Rightarrow \frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = \lambda$  (Separation Constant)  $\Rightarrow$  可以使用此方法!

1.Homo, 假设:  $u(0, t) = u(L, t) = 0; u(x, 0) = f(x)$     2.Inhomo, 假设: change to  $u(0, t) = a, u(L, t) = b$

**Basic Idea|Homo:** 1. 假设解  $u$  能写成  $u(x, t) = X(x)T(t) \Rightarrow$  2. 代入 PDE, 得两个 ODE  $\Rightarrow$  3. 解其中一个得到特征函数  $X_n \Rightarrow$  4. 代 initial 条件得限制条件  $\Rightarrow$  5. 解另一个得到特征函数  $T_n \Rightarrow$  6. 特征函数  $X_n T_n$  linear combination 得到  $u$ . 7. 代 boundary 条件用 Euler-Fourier 得到的系数.

· Remark:  $^1$  化简中可以使用正比简化计算     $^2$  对  $n$  的取值可为  $1, 2, 3, \dots$      $^3$  对  $\lambda$  的取值让出现  $\sin, \cos$  时为正不容易出错, 或注意正负号

**Basic Idea|Inhomo:** Soln:  $u(x, t) = v(x) + \omega(x, t), v(x) = \frac{b-a}{L}x + a$  ||  $\omega(x, t)$  当作 homo 微分方程解, 边界条件  $\omega(x, 0) = f(x) - v(x); \omega(0, t) = \omega(L, t) = 0$

**Laplace Equation:** 2D:  $\nabla^2 u = \partial_x^2 u + \partial_y^2 u = 0$     3D:  $\nabla^2 u = \partial_x^2 u + \partial_y^2 u + \partial_z^2 u = 0$     Following are the method to solve:

1. **Rectangle:**  $\{0 \leq x \leq a, 0 \leq y \leq b\}$  with  $u(x, 0) = u(x, b) = u(0, y) = 0; u(a, y) = f(y)$     直接按照 Homo 正常计算     $\Downarrow$  可能用到 Euler Equation

2. **Circle:**  $\{u(r = a, \theta) = f(\theta); u(r, \theta) \text{ bounded}\}$      $^1 \partial_x^2 u + \partial_y^2 u = 0; z = r \cos(\theta), y = r \sin(\theta)$  变极坐标  $R(r), \Theta(\theta)$      $^2$  一样的计算 + 条件  $\Theta(\theta + 2\pi) = \Theta(\theta)$

6 Sturm-Liouville Theory (Another 'Fourier')

**6.1 Sturm-Liouville Theory**     $P_y y'' + Q_y y' + R_y y = 0 \Rightarrow [\mu P_y y']' + \mu R_y y = 0$  Integrating factor:  $\mu = \frac{1}{p} e^{\int_{x_0}^x \frac{Q(s)}{P(s)} ds}$

**Sturm-Liouville (SL) Problem:** Form:  $-(p(x)y')' + q(x)y = \lambda r(x)y$  ,  $p, r > 0$  || With:  $a_1 y(a) + a_2 y'(a) = b_1 y(b) + b_2 y'(b) = 0$      $p, p', q, r$  continuous. || **weight function:**  $r(x)$ . ||  $\Downarrow$  **Lagrange's Identity**

·  $\Rightarrow$  Let *Linear Operator:*  $\mathcal{L}(y) := -(p(x)y')' + q(x)y \Rightarrow$  Problem becomes:  $\mathcal{L}(y) = \lambda r(x)y$  (It's a Eigenvalue Problem  $\mathcal{L}(\Psi_n) = \lambda_n r(x) \Psi_n$ )  $\Rightarrow \forall$  solutions  $u, v$  with (二阶导连续) of SL Problem:  $\langle \mathcal{L}(u), v \rangle = \langle u, \mathcal{L}(v) \rangle$  !!!

· **Properties|Real|Orthogonal:** In SL Problem:  $^1 \lambda_n, \Psi_n$  are *real*; ||  $^2$  Eigenfunctions  $\Psi_1, \Psi_2$  with  $\lambda_1 \neq \lambda_2$ . **Modified Inner Product:**  $\langle \Psi_1, \Psi_2 \rangle = \int_a^b r(x) \Psi_1 \Psi_2 dx = 0$  i.e.  $\Psi_1, \Psi_2$  orthogonal.

· **Properties|Simple|Ordered:** In SL Problem:  $^3 \lambda_n$  are *simple*:  $\forall \lambda_n$  has *unique*  $\Psi_n$ ; ||  $^4 \lambda_n$  are *Ordered*:  $\lambda_1 < \lambda_2 < \dots < \lambda_n < \dots \rightarrow \infty$  as  $n \rightarrow \infty$

**Normalisation of Eigenfunction:**  $^1 \Psi_n$  normalised if:  $\langle \Psi_n, \Psi_n \rangle = 1$  ||  $^2$  Method (Normalised  $\Psi_n$ ):  $\Phi_n = \frac{1}{\sqrt{\langle \Psi_n, \Psi_n \rangle}} \Psi_n = [\int_a^b r(x) \Psi_n^2 dx]^{-1/2} \Psi_n$

**Sturm-Liouville Convergence Theorem:** If  $f, f'$  are piecewise continuous on  $[0, 1] \Rightarrow F(x) = \sum_{n=1}^{\infty} c_n \Phi_n$  converges. ||  $\Phi_n$  are *orthonormal* eigenfunction of a Sturm-Liouville problem.

· Furthermore,  $^1 F \rightarrow f$  when  $f$  is continuous;     $^2 F \rightarrow \frac{f(x^+) + f(x^-)}{2}$  when  $f$  is discontinuous.    边界条件右边界条件决定    **Calculate Coefficients:**  $c_n = \langle f, \Phi_n \rangle = \int_a^b r(x) f(x) \Phi_n(x) dx$  || ps:  $c_n$  minimize the MSE  $\epsilon_N$

· 如果我们不需要归一化: 特征函数  $\Psi_n \Rightarrow f(x) = \sum_{n=1}^{\infty} c_n \Psi_n$  ,  $c_n = \langle f, \Psi_n \rangle = \int_a^b f(x) \Psi_n(x) dx$  || **Parseval's Identity:**  $f, f'$  are piecewise continuous on  $[a, b] \Rightarrow \langle f, f \rangle = \int_a^b f^2(x) r(x) dx = \sum_{n=1}^{\infty} c_n^2$

**6.2 General Boundary eigenvalue problem || Non-Homogeneous Boundary Value Problems || Open Interval Situation**

**General Boundary eigenvalue problem:**  $\mathcal{L}(y) = \lambda r(x)y$  ,  $a < x < b$  || With  $n - th$  order operator:  $\mathcal{L}(y) = p_n(x) \frac{d^n y}{dx^n} + \dots + p_1(x) \frac{dy}{dx} + p_0(x)y$  || Assume  $n$  linear Homogeneous Boundary conditions at the endpoints.

$\Rightarrow$  And suppose that  $\langle \mathcal{L}(u), v \rangle = \langle u, \mathcal{L}(v) \rangle \Leftarrow (\text{Self-Adjoint Operator}) \rightarrow \mathcal{L}$  must be even order. || **Properties:**  $^1$  Infinite eigenvalues.     $^2$  Orthogonal eigenfunctions.     $^3$  类似 Fourier 可以将任意函数用级数展开.

**Non-Homogeneous BVP|Sturm-Liouville:**  $\mathcal{L}(y) = \mu r(x)y + f(x)$     where  $^1 \mathcal{L}$  is Sturm-Liouville operator. (就是符合前面 SL Problem operator 的定义 + conditions)     $^2 \mu$  is constant.

·  $\Rightarrow$  Let  $\lambda_n, \Phi_n$  be eigenvalues & orthonormal eigenfunctions of  $\mathcal{L}(y) = \lambda r(x)y$ . || Method: **Green's Function:**  $G(x, z) = \sum_{n=1}^{\infty} \frac{\Phi_n(x) \Phi_n(z)}{\lambda_n - \mu}$  ( $\mu \neq \lambda_n$ )  $\Rightarrow$  Solution:  $y(x) = \int_a^b G(x, z) f(z) dz$  ( $\mu \neq \lambda_n$ )

· (Ctd.)  $^1$  If  $\mu \neq \lambda_n \Rightarrow$  unique solution.     $^2$  If  $\exists m, \mu = \lambda_m$ : Let  $c_n = \int_a^b f(x) \Phi_n(x) dx \Rightarrow$  Case I ( $c_m = 0$ ): Let  $b_n = \frac{c_n}{\lambda_n - \mu}, n \neq m$      $y(x) = \sum_{n=1, n \neq m}^{\infty} [b_n \Phi_n(x)] + b_m \Phi_m(x), b_m \in \mathbb{R}$     Case II ( $c_m \neq 0$ ): No Solution.

**Singular / Regular SL Problem:** boundary condition at *closed interval*. || **Singular:** boundary condition at *open interval* or [a,b) or (a,b].

**Signlar / SL Problem|One-side:**  $\mathcal{L}(y) = \lambda r(x)y$ , 奇异边  $x = a, b$ .  $\Rightarrow \lim_{\epsilon \rightarrow a,b} p(\epsilon) [u'(\epsilon) v(\epsilon) - u(\epsilon) v'(\epsilon)] = 0 \Rightarrow \forall$  sols  $u, v$ :  $\int_a^b \mathcal{L}(u) v - u \mathcal{L}(v) = 0 \star \Leftarrow \lim_{\epsilon \rightarrow a,b} p(\epsilon) = 0$  and  $u, u', v, v'$  bounded as  $x \rightarrow a, b$

·  $\star$  holds  $\Rightarrow$   $^1$  特征值实数  $^2$  正交特征函数  $^3$  可以展开任意函数 || 解题一般思路:  $^1$  分离变量法得到 ODE  $^2$  解 ODE(观察是否是 Bessel 问题), 系数先不管  $^3$  使用边界条件 (注意 bounded) 去除部分无关系数  $^4$  通解: 特征函数线性组合 + series 方法确定系数

**6.3 Other Results on Sturm-Liouville Theory**

**Mean Square Error:**  $\epsilon_N = \int_a^b r(x) [f(x) - S_N(x)]^2 dx$  (Truncation)  $S_N(x) = \sum_{n=1}^N c_n \Phi_n(x)$  || If  $\lim_{N \rightarrow \infty} \epsilon_N = 0 \Rightarrow F = \sum_{n=1}^{\infty} c_n \Phi_n$  **converge in the mean** to  $f$ . (Not Pointwise Convergence)

**Complete of Eigenfunctions:**  $\{ \Phi_n \}$  (or  $\{ \Psi_n \}$ ) is complete with respect to [Mean Square Convergence] / [Pointwise Convergence] for a class of functions  $\mathcal{F}$     if  $[ \lim_{N \rightarrow \infty} \epsilon_N = 0 ] / [ \lim_{N \rightarrow \infty} S_N(x) = f(x) ]$  for all  $f \in \mathcal{F}$

**Class of Square Integrable functions:**  $\mathcal{F} = \{ f(x) | f \text{ and } f^2 \text{ are integrable on } x \in [a, b] \}$

**Theorem|Completeness:** The Set orthonormal eigenfunctions  $\{ \Phi_n \}$  of a regular SL boundary problem are *complete* with respect to [Mean Square Convergence]. for a *class* of Square Integrable functions.

7 Laplace's Transform

**Laplace Transform:** The Laplace transform of  $f$  is:  $F(s) = \mathcal{L}\{f(t)\}(s) = \int_0^{\infty} e^{-st} f(t) dt$  || ps: May not exists

$f \in E$  **xponential Order:**  $^1 f$  is piecewise continuous on  $[0, \infty)$ ;  $^2 \forall t \geq 0 : \exists A, B > 0$  s.t.  $|f(t)| \leq A e^{Bt}$

**Theorem|Laplace Existence:** If  $f \in E \Rightarrow \forall s$  sufficiently large,  $\mathcal{L}\{f(s)\}$  exists.

**Theorem|Linear Operator:**  $\mathcal{L}\{c_1 f + c_2 g\}(s) = c_1 \mathcal{L}\{f\}(s) + c_2 \mathcal{L}\{g\}(s)$      $\mathcal{L}(0) = 0$

**Theorem|Derivative:** If  $f$  continuous on  $[0, \infty)$ ;  $f, f' \in E \Rightarrow \mathcal{L}\{f'\}(s) = s \mathcal{L}\{f\}(s) - f(0)$

·  $f, f', \dots, f^{(n-1)} \in E$  and continuous  $\Rightarrow \mathcal{L}\{f^{(n)}\}(s) = s^n \mathcal{L}\{f\}(s) - s^{n-1} f(0) - \dots - s f^{(n-2)}(0) - s^0 f^{(n-1)}(0)$

**Inverse Laplace Transform:**  $f(t) = \mathcal{L}^{-1}\{F(s)\}(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} F(s) ds$  ,  $\Re(s) = \gamma$  (general form)

| $f(t)$              | $\mathcal{L}\{f\}(s)$         | $f(t)$                | $\mathcal{L}\{f\}(s)$                               | $f(t)$                | $\mathcal{L}\{f\}(s)$                                 |
|---------------------|-------------------------------|-----------------------|-----------------------------------------------------|-----------------------|-------------------------------------------------------|
| 1                   | $\frac{1}{s}$                 | $t^n$   $e^{at}$      | $\frac{n!}{s^{n+1}}$   $\frac{1}{s-a}, x > a$       | $t^n e^{at}$          | $\frac{n!}{(s-a)^{n+1}}$                              |
| $\ominus \sinh(at)$ | $\frac{s^2 - a^2}{s^2 + a^2}$ | $\sin(at)$            | $\frac{a}{s^2 + a^2}$                               | $\cos(at)$            | $\frac{s}{s^2 + a^2}$                                 |
| $\ominus \cosh(at)$ | $\frac{s^2 - a^2}{s^2 + a^2}$ | $e^{at} \sin(bt)$     | $\frac{(s-a)^2 + b^2}{(s-a)^2 + b^2}$               | $e^{at} \cos(bt)$     | $\frac{s-a}{(s-a)^2 + b^2}$                           |
| $u_c(t)$            | $\frac{e^{-sc}}{s}$           | $t^n e^{at} \sin(bt)$ | $(-1)^n \frac{d^n}{ds^n} (\frac{b}{(s-a)^2 + b^2})$ | $t^n e^{at} \cos(bt)$ | $(-1)^n \frac{d^n}{ds^n} (\frac{s-a}{(s-a)^2 + b^2})$ |

ps: 一般:  $s > 0$  ||  $\sinh, \cosh: s > |a|$  ||  $e^{at} \sin/\cos(t^n): s > a$  ||  $\star$  **Heaviside Unit Function:**  $u_c = \mathcal{X}(c, \infty)$

**Uniqueness & Existence:**  $^1$  If  $f, g$  continuous  $\Rightarrow \mathcal{L}\{f\} = \mathcal{L}\{g\} \Leftrightarrow f = g$

$^2$  If  $f, g$  pointwise continuous  $\Rightarrow$  唯一性不一定成立 (非连续点上不同) (No practical significance in applications)

8 Appendix

**Wronskian Geometric Meaning:** Geometrically,  $W$  is the volume spanned by the three vectors  $y^{(1)}, y^{(2)}, y^{(3)}$ ;  $t$ : 增加或减小代表在  $y^{(i)}$  方向上 compress/stretch.

和差化积/积化和差:  $s+s=sc, s-s=cs, c+c=cc, c-c=ss$  || e.g.  $\sin A + \sin B = 2 \sin(\frac{A+B}{2}) \cos(\frac{A-B}{2})$      $\sin A \sin B = -\frac{1}{2} [\cos(A+B) - \cos(A-B)]$  ||  $\sin(2x) = 2 \sin(x) \cos(x)$      $\cos(2x) = 2 \cos^2(x) - 1$      $\cos(x) = \sqrt{\frac{1+\cos(2x)}{2}}$      $\sin(x) = \sqrt{\frac{1-\cos(2x)}{2}}$

**Hyperbolic Func:**  $\sinh(x) = \frac{e^x - e^{-x}}{2}$      $\cosh(x) = \frac{e^x + e^{-x}}{2}$      $\cosh^2(x) - \sinh^2(x) = 1$      $[\sinh(x)]' = \cosh(x)$      $[\cosh(x)]' = \sinh(x)$  ||  $\sinh(x \pm y) = \sinh(x) \cosh(y) \pm \cosh(x) \sinh(y)$      $\cosh(x \pm y) = \cosh(x) \cosh(y) \pm \sinh(x) \sinh(y)$

·  $\sinh(x)$  单调递增, odd, iff  $x > 0, \sinh(x) > 0$      $\cosh(x) > 0$  单调递增, even ||  $\sinh(2x) = 2 \sinh(x) \cosh(x)$      $\cosh(2x) = \cosh^2(x) + \sinh^2(x)$

**Common Int:**  $\int \tan(x) dx = -\ln|\cos(x)|$ ;  $\int \cot(x) dx = \ln|\sin(x)|$ ;  $\int \sec(x) dx = \ln|\sec(x) + \tan(x)|$ ;  $\int \csc(x) dx = \ln|\csc(x) - \cot(x)| = \ln|\tan(\frac{x}{2})|$

**Common Der:**  $[\tan(x)]' = \sec^2(x)$ ;  $[\cot(x)]' = -\csc^2(x)$ ;  $[\sec(x)]' = \sec(x) \tan(x)$ ;  $[\csc(x)]' = -\csc(x) \cot(x)$ ;  $[\arctan(x)]' = \frac{1}{1+x^2}$ . ||  $1 + \tan^2(x) = \sec^2(x)$ ;  $1 - \csc^2(x) = \cot^2(x)$ ;  $\csc^2(x) - \cot^2(x) = 1$ .

**Polar Coordinate:**  $x = r \cos(\theta), y = r \sin(\theta)$     we have:  $r \dot{r} = x \dot{x} + y \dot{y}$  ,  $y \dot{x} - x \dot{y} = -r^2 \dot{\theta}$

**Explain of Fourier Series:** In application, it's decomposition of signal into infinite (countable) waves; each wave  $\cos(\frac{n\pi x}{L}), \sin(\frac{n\pi x}{L})$  has discrete period;  $a_n, b_m$  correspond to amplitudes of waves. || In vector algebra, it's a space of periodic functions;  $f$  corresponds to an arbitrary vector  $v$ ; orthogonal basis  $\{1, \cos(\frac{n\pi x}{L}), \sin(\frac{n\pi x}{L})\}$ ;  $a_n, b_m$  correspond to coefficient of each components of basis.

**Even & Odd Function:** Even:  $f(-x) = f(x)$ ; Odd:  $f(-x) = -f(x)$ . || Property:  $^1 f(x)$  odd:  $\int_{-L}^L f(x) dx = 0$      $^2 f(x)$  even:  $\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$  ||  $^1$  eve + eve = eve;  $^2$  odd + odd = odd;  $^3$  eve x eve = eve;  $^4$  odd x odd = eve;  $^5$  eve x odd = odd.

**Euler Formula:**  $e^{ix} = \cos(x) + i \sin(x)$  ||  $e^{inx} = \cos(nx) + i \sin(nx)$  ||  $\cos(nx) = \frac{e^{inx} + e^{-inx}}{2}$  ||  $\sin(nx) = \frac{e^{inx} - e^{-inx}}{2i}$

**Solve  $y'' + \lambda y = 0$ :**  $^1 \lambda < 0 : y = c_1 \sinh(\sqrt{-\lambda} x) + c_2 \cosh(\sqrt{-\lambda} x)$      $^2 \lambda = 0 : y = c_1 x + c_2$      $^3 \lambda > 0 (C) : y = c_1 \cos(\sqrt{\lambda} x) + c_2 \sin(\sqrt{\lambda} x)$  ||    **Solve  $y'' + \beta y = 0$ :**  $y = c_1 e^{-\beta x}$

**Approximation  $\sin(x) \approx x$  when  $x \rightarrow 0$**  ||  $\Downarrow$  一般对于  $-(xy')' = \lambda xy$     let  $t = \sqrt{\lambda} x \Rightarrow y'' + \frac{1}{t^2} y' + y = 0$

**Bessel's Equation:**  $y'' + \frac{1}{t} y' + (1 - \frac{m^2}{t^2}) y = 0 \Rightarrow$  Solution:  $y(t) = c_1 J_m(t) + c_2 Y_m(t)$  ||  $J_m(t)$  is Bessel function of first kind;  $Y_m(t)$  is Bessel function of second kind. || ps:  $Y_m(t)$  not bounded as  $t \rightarrow 0, J_m(t)$  bounded.

**8.1 Some Applications**

**Generalised Heat Equation:**  $r(x) \partial_t u = \partial_x [p \partial_x u] - qu + F \Rightarrow$   $^1$  考虑 Homo: 分离变量, 得到一个 SL 问题; 一个简单 ODE. ODE 不计算.  $^2$  将  $F/r$  进行 SL 分解为 Series.  $^3$  考虑解  $u = \sum X_n(x) b_n(t), b_n$  未知;  $^2, 3$  步入原始方程, 根据特征向量正交得系数关系: 一个 ODE.  $^4$  求解 ODE 系数由其他边界条件给出.